MÒDUL D’AUTOMÒBIL: RETROVISORS

Posar en pràctica els coneixements adquirits durant el curs per a desenvolupar un sistema electrònic seguint el flux de disseny de PCBs amb temàtica de projectes d’automoció.

Grau d’Enginyeria Electrònica de Telecomunicacions: Facultat de Física- UB

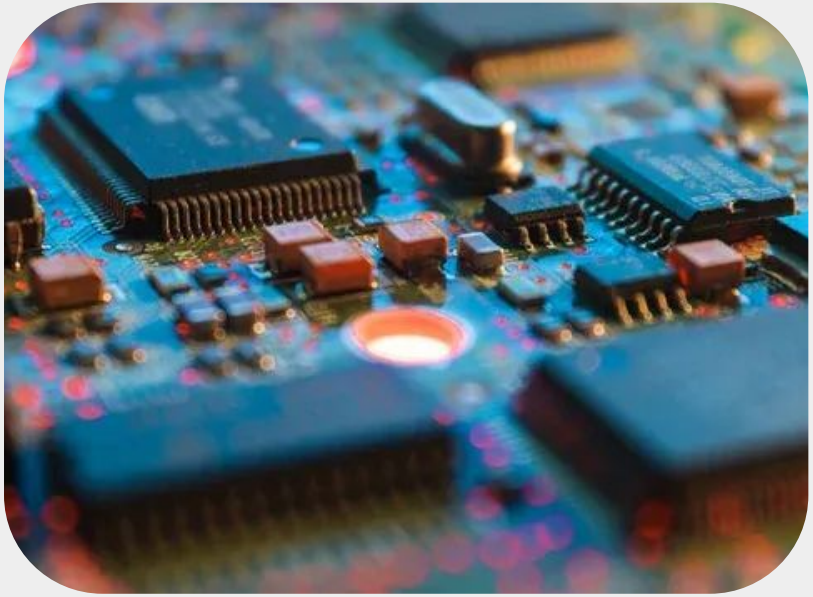
Eines de Disseny - 2026

Carlos Soler Fernandez, Ingrid Lopez Sanchez

UNIVERSITAT DE BARCELONA

Objectius principals:

- ✓ Dissenyar i fabricar un sistema electrònic integral per al control de retrovisors
- ✓ Gestionar els tres motors amb els finals de carrera on calgui
- ✓ Incorporar un sistema de calefacció i sensors digitals d'angle mort
- ✓ Assegurant la viabilitat tècnica i econòmica des del prototip fins a la producció a gran escala



DATASHEETS

PIC24HJ128GPX02

Microcontrolador: Dispositiu encarregat de controlar el funcionament de la placa. Funciona a 3V aprox i 200mA.

TSR0.5-2450

Regulador de tensió: Dispositiu utilitzat per passar de 12V (Vpwr) a 5V .

ABLS2-8MHz-D4Y-T

Clock/Oscil·lador: Proporciona al PIC una senyal de rellotge constant per executar cada línia i regula la freqüència del senyal. Funciona a 8MHz

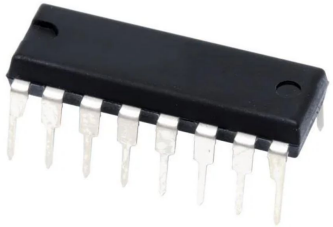
MCP2551

Bus CAN (Controller Area Network): connecta el mòdul de retrovisors amb la resta del cotxe.

DATASHEETS

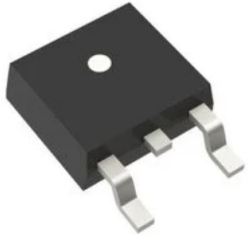
PCF8574

Expansor I2C: Expansor d'entrada/sortida de 8-bits (I/O) per a un bus bidireccional de dues línies (I2C)



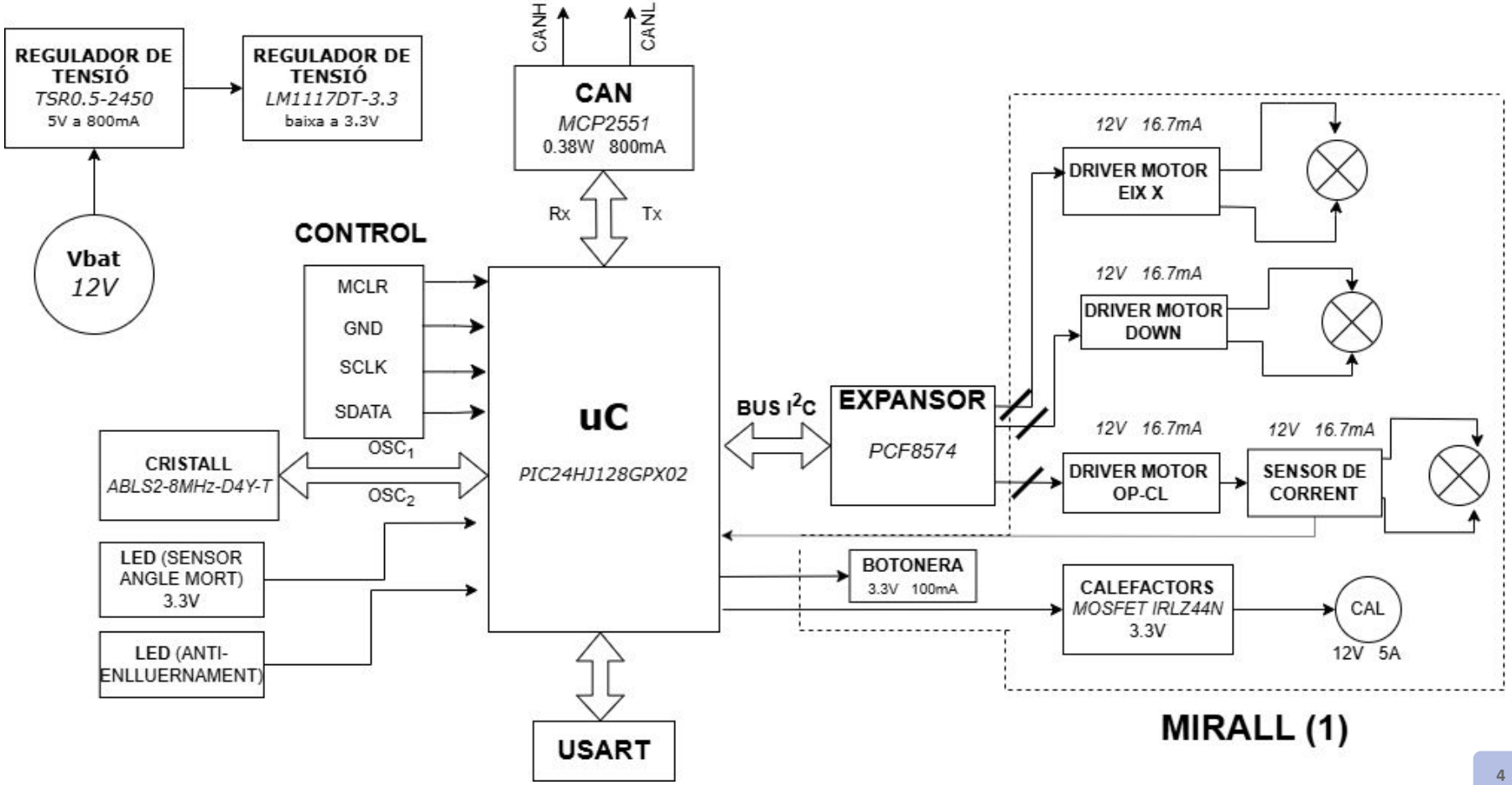
LM1117DT-3.3

Regulador de tensió (LDO): Dispositiu utilitzat per passar de 5V () a 3.3V (Vcc) i mantenir-los constants.

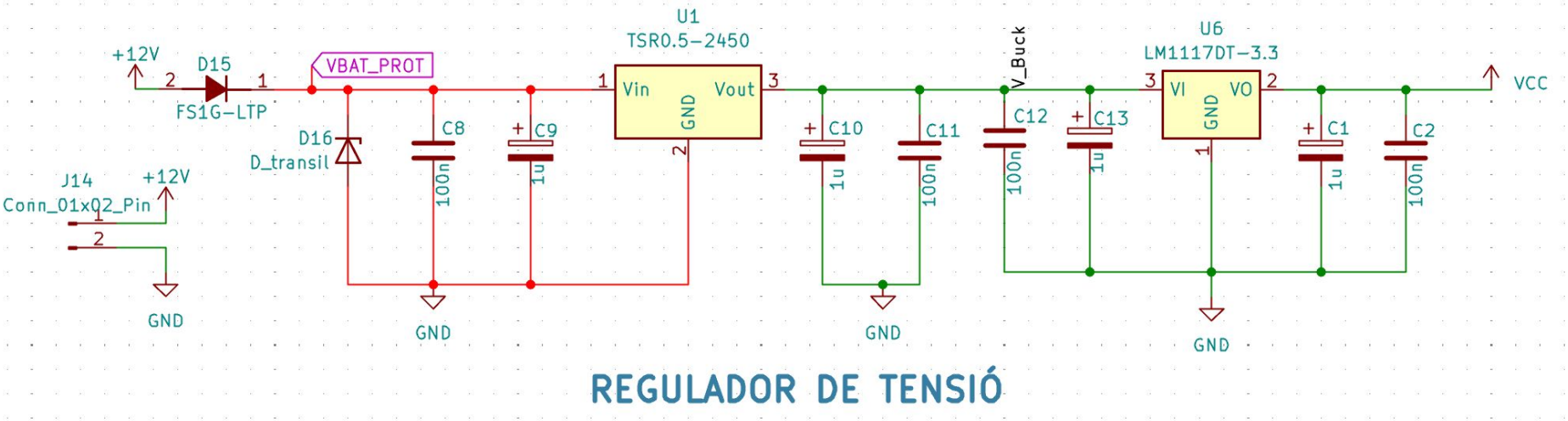


G6E

Relé: Dispositiu utilitzat als drivers de motor per tal de presentar un pont en H.

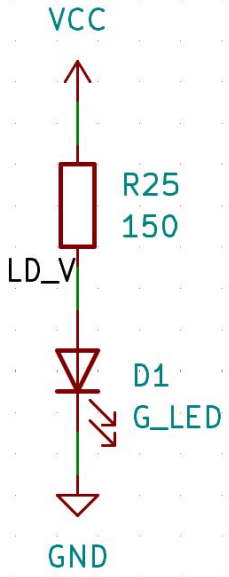

REGULADOR DE TENSIÓ
TSR0.5-2450
5V a 800mA
REGULADOR DE TENSIÓ
LM1117DT-3.3
baixa a 3.3V
Vbat
12V
CANH
CANL
CAN
MCP2551
0.38W 800mA
Rx
Tx
CONTROL
MCLR
GND
SCLK
SDATA
OSC1
OSC2
CRISTALL
ABLS2-8MHz-D4Y-T
LED (SENSOR ANGLE MORT)
3.3V
LED (ANTI-ENLLUERNAMENT)
uC
PIC24HJ128GPX02
USART
BUS I2C
EXPANSOR
PCF8574
12V 16.7mA
DRIVER MOTOR EIX X
12V 16.7mA
DRIVER MOTOR DOWN
12V 16.7mA
DRIVER MOTOR OP-CL
12V 16.7mA
SENSOR DE CORRENT
BOTONERA
3.3V 100mA
CALEFACTORS
MOSFET IRLZ44N
3.3V
CAL
12V 5A
MIRALL (1)

REGULADOR DE TENSIÓ

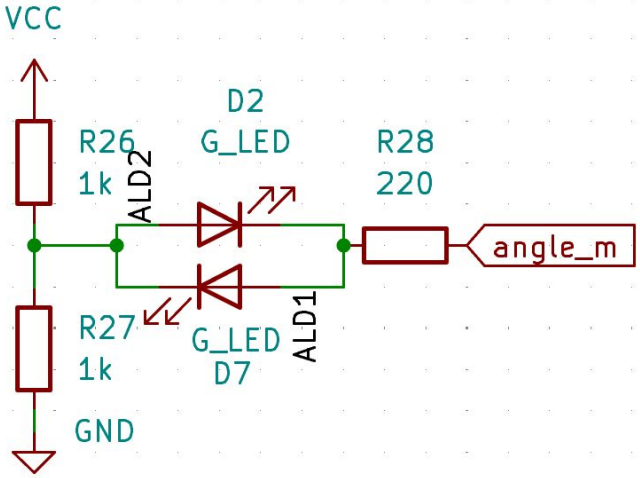


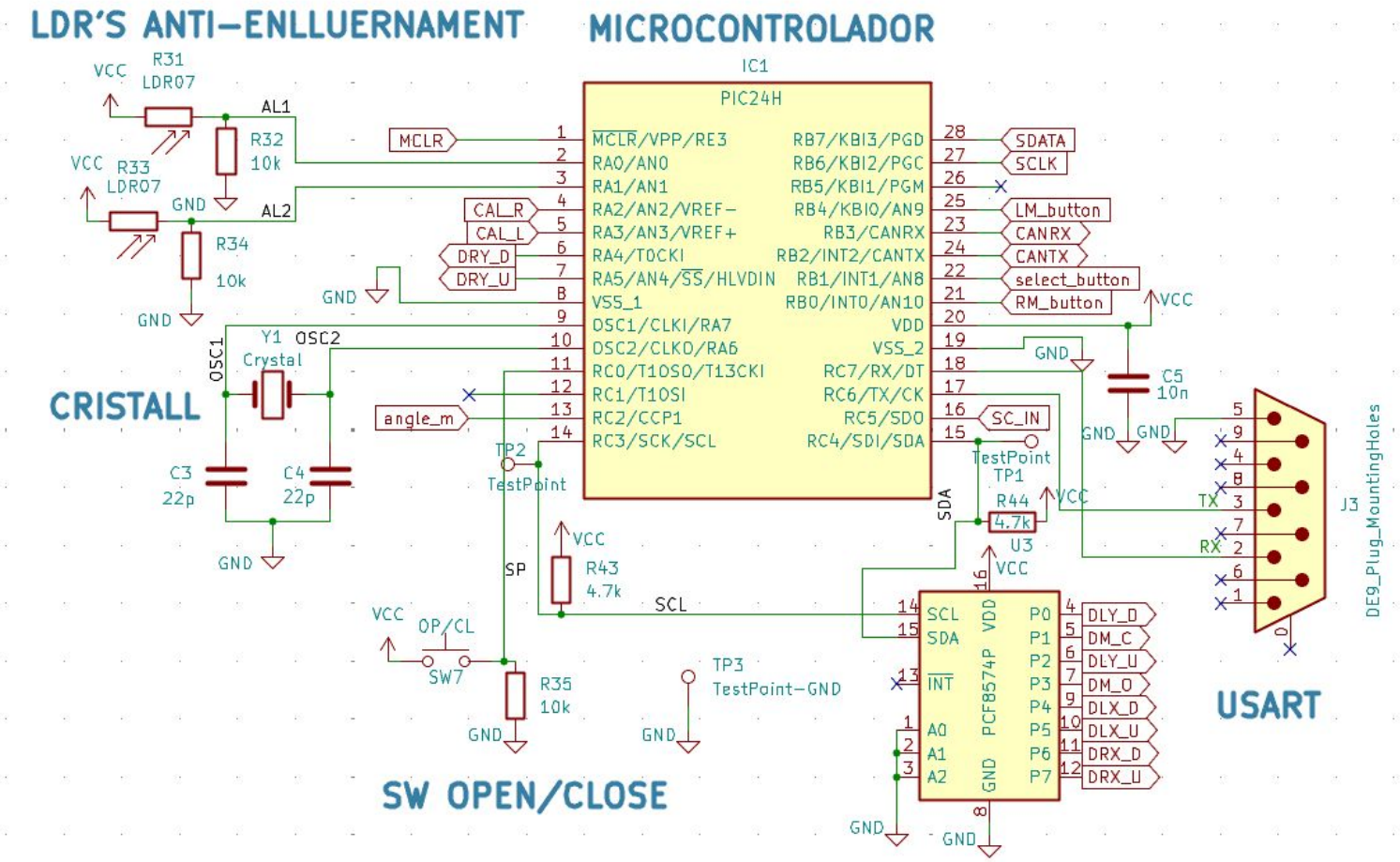
LED'S

Led per a verificar que hi ha corrent.

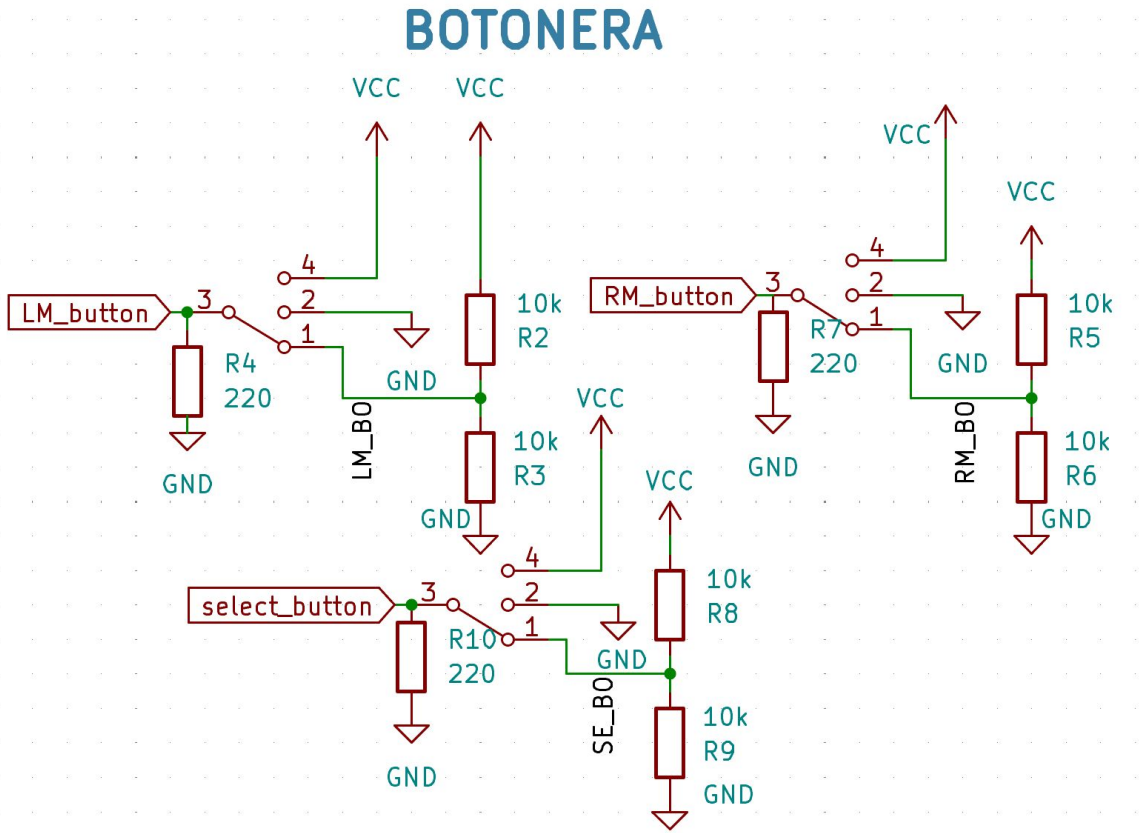


LEDS SENSOR ANGLE MORT




LDR'S ANTI-ENLLUERNAMENT
MICROCONTROLADOR
IC1
PIC24H
CRISTALL
SW OPEN/CLOSE
USART
J3
DE9_Plug_MountingHoles
U3
PCF8574P
R31 LDR07
R32 10k
R33 LDR07
R34 10k
Y1 Crystal
C3 22p
C4 22p
C5 10n
R43 4.7k
R44 4.7k
R35 10k
SW7
OP/CL
TP1 TestPoint
TP2 TestPoint
TP3 TestPoint-GND
VCC
GND
AL1
AL2
OSC1
OSC2
SP
SCL
SDA
TX
RX
MCLR
CAL_R
CAL_L
DRY_D
DRY_U
angle_m
SDATA
SCLK
LM_button
CANRX
CANTX
select_button
RM_button
SC_IN
DLY_D
DM_C
DLY_U
DM_O
DLX_D
DLX_U
DRX_D
DRX_U

FASE DIGITAL



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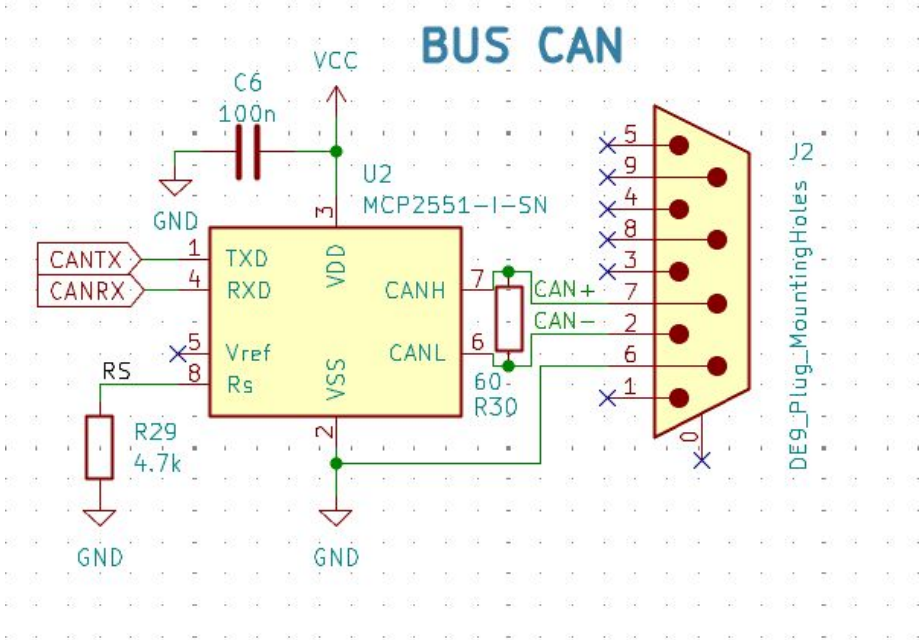
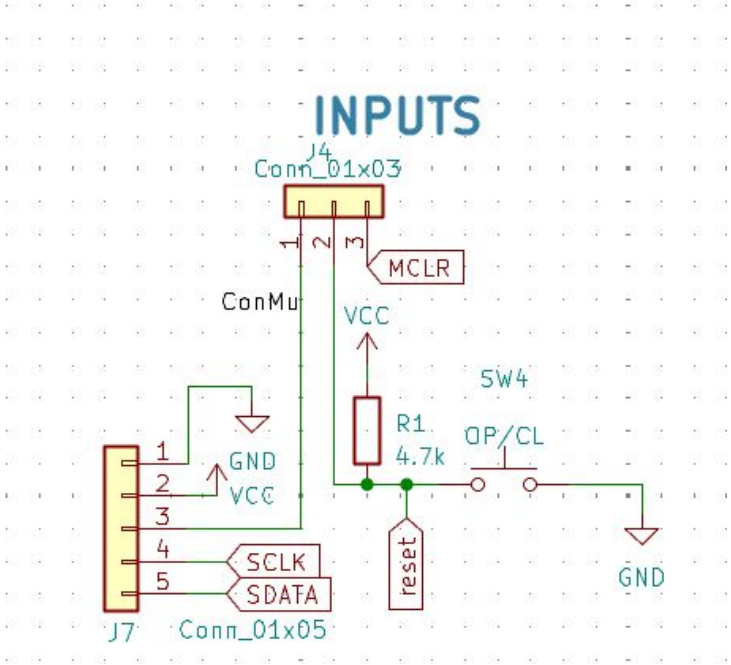
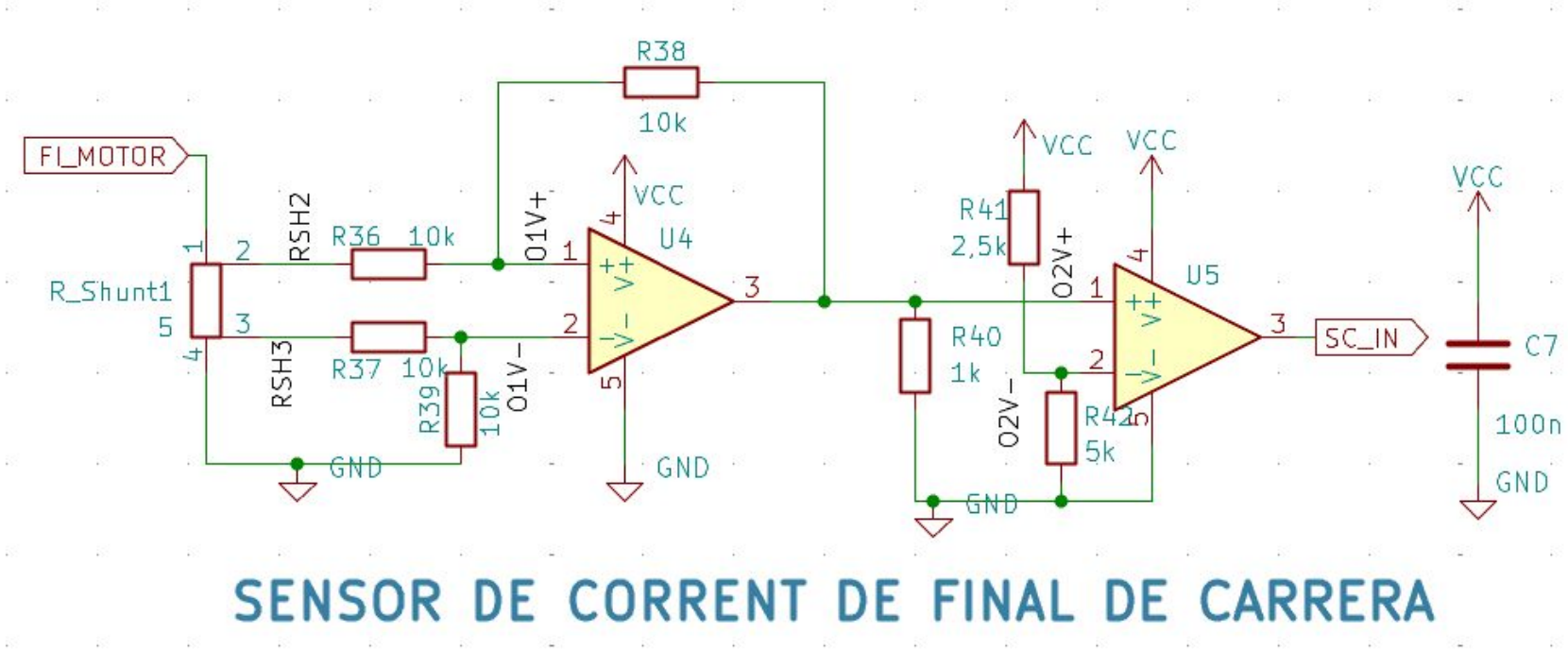
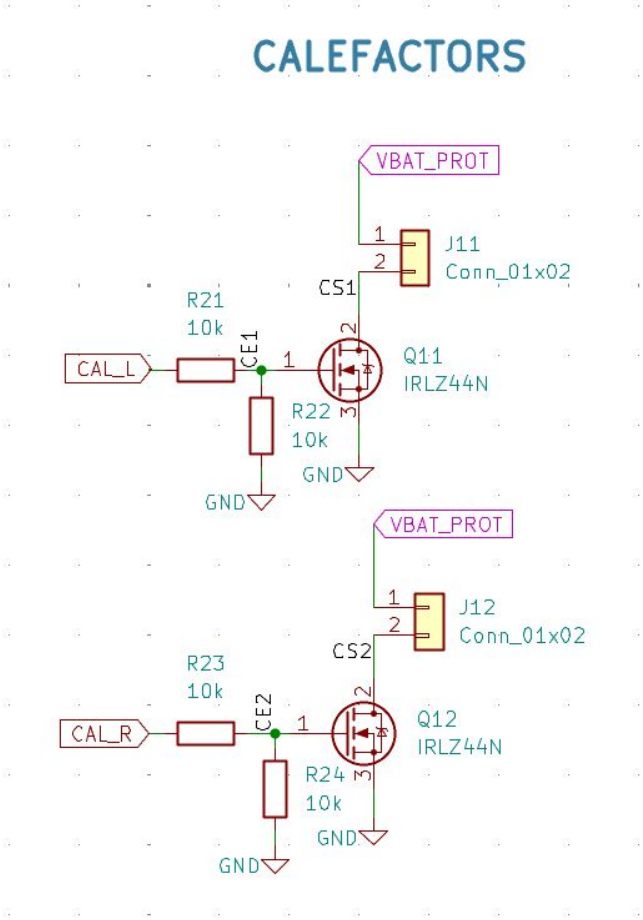




FASE DE POTÈNCIA



FASE DE POTÈNCIA



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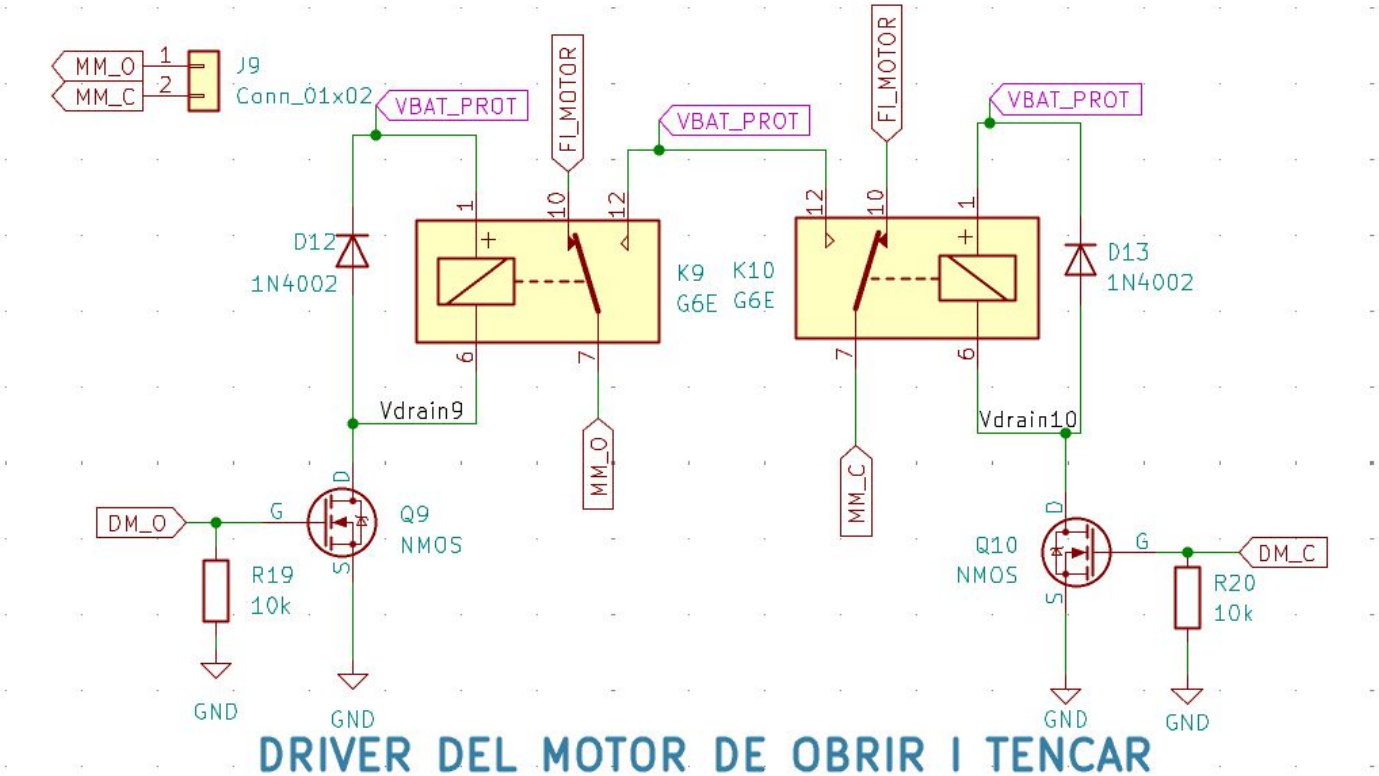

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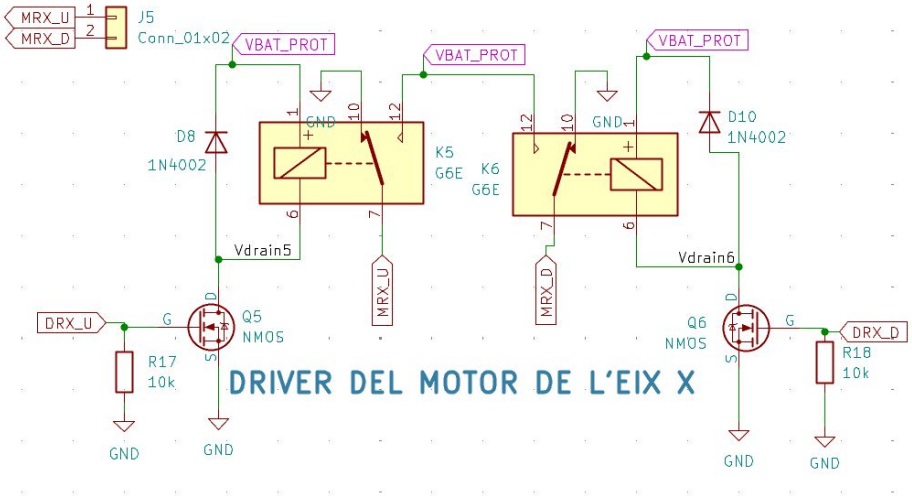
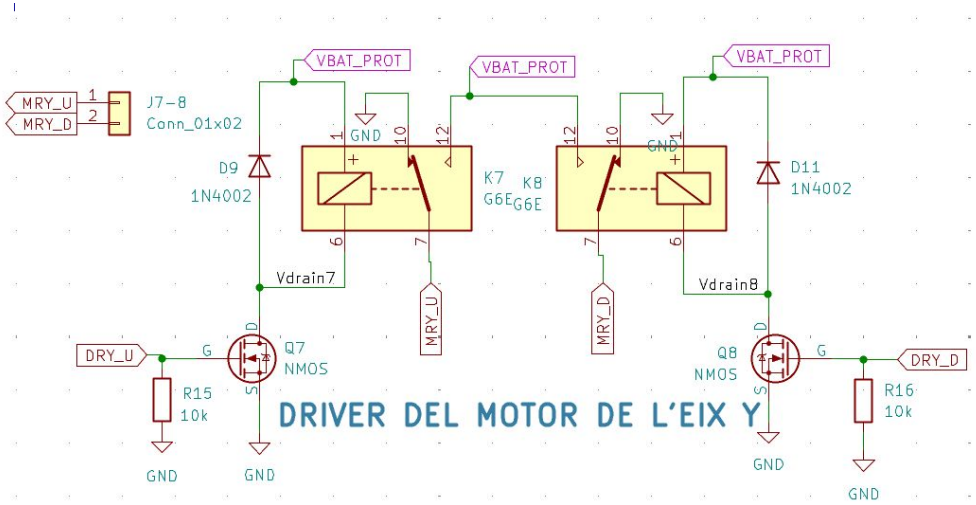
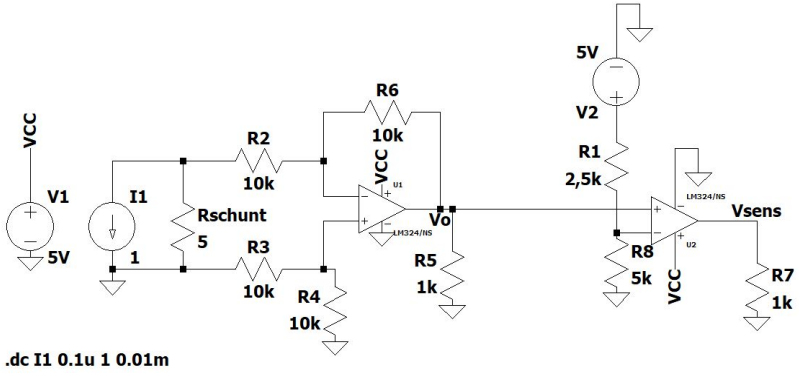
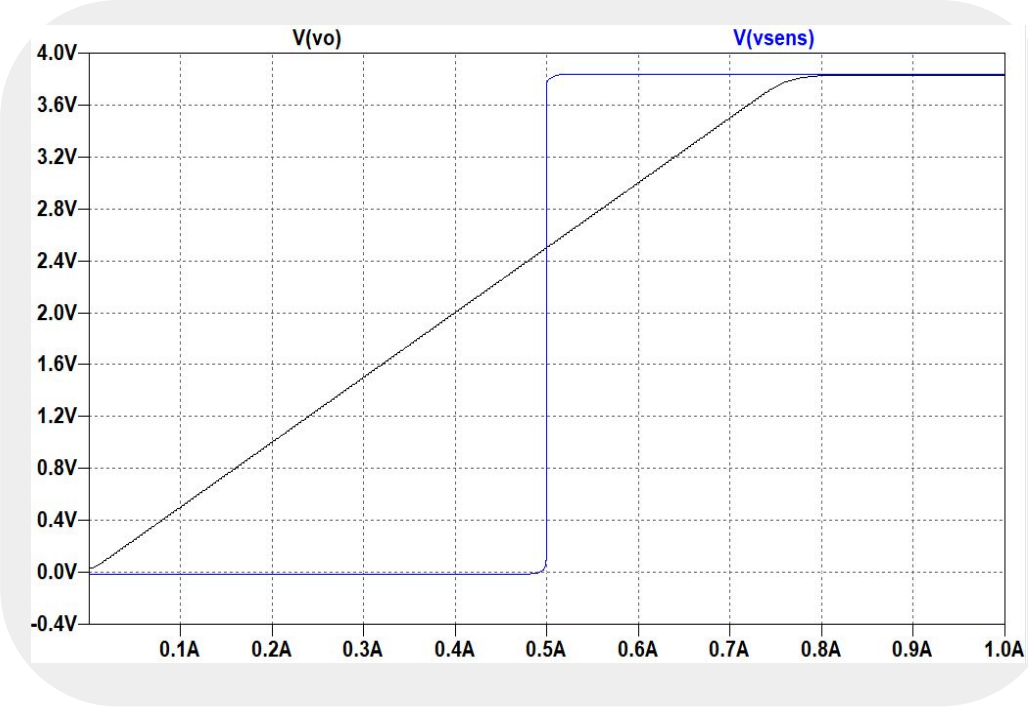




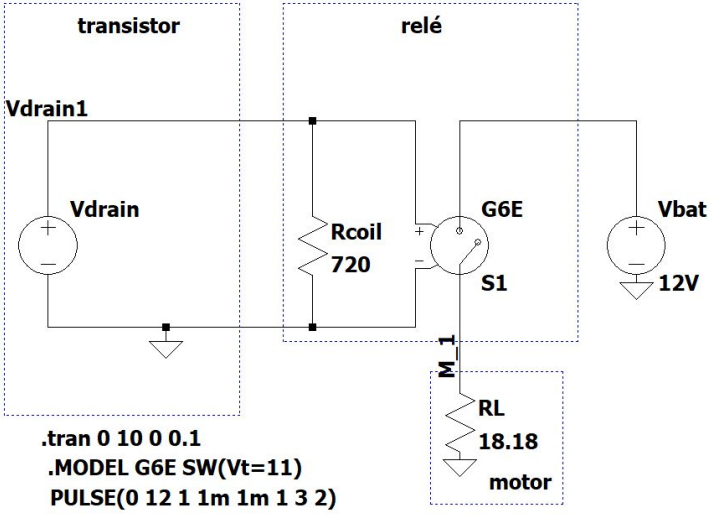
Al haber fet servir els ponts en H, l'estructura de tots els drivers és la mateixa, tenint 5 ponts en H en total: dos per moure en eix X, dos per moure en eix Y (un per cada retrovisor) i un per obrir/tencar.

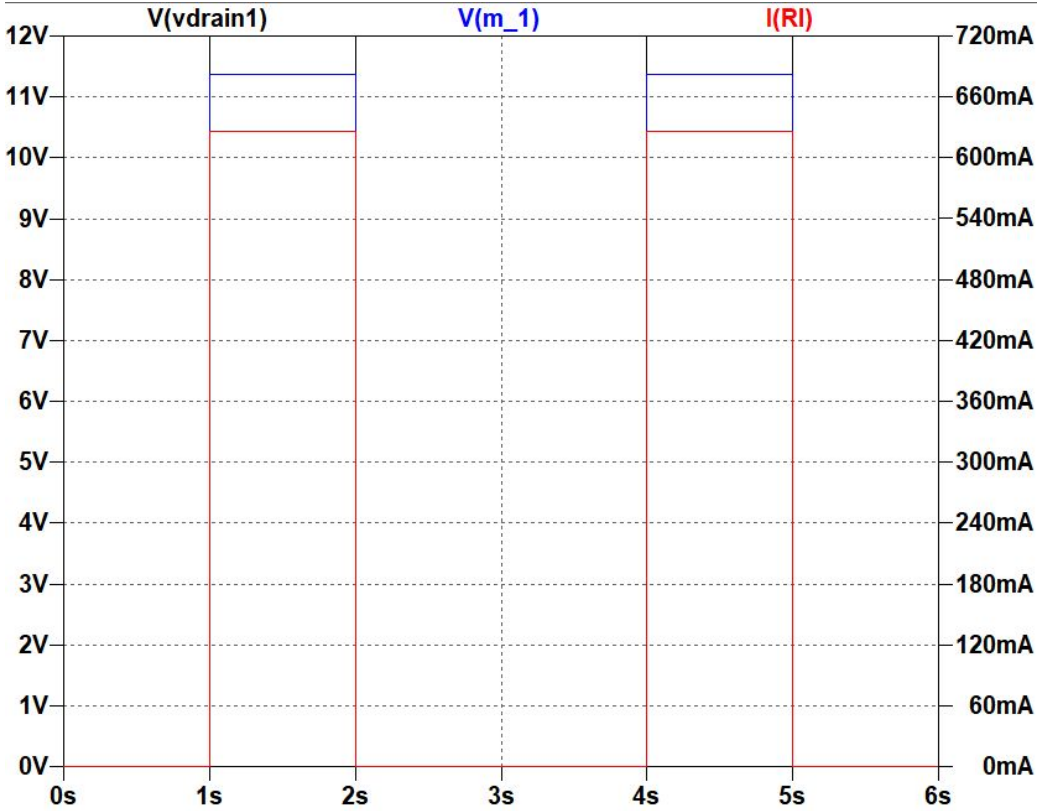
SIMULACIÓ: SENSOR DE CORRENT; FINAL DE CARRERA OP/CL



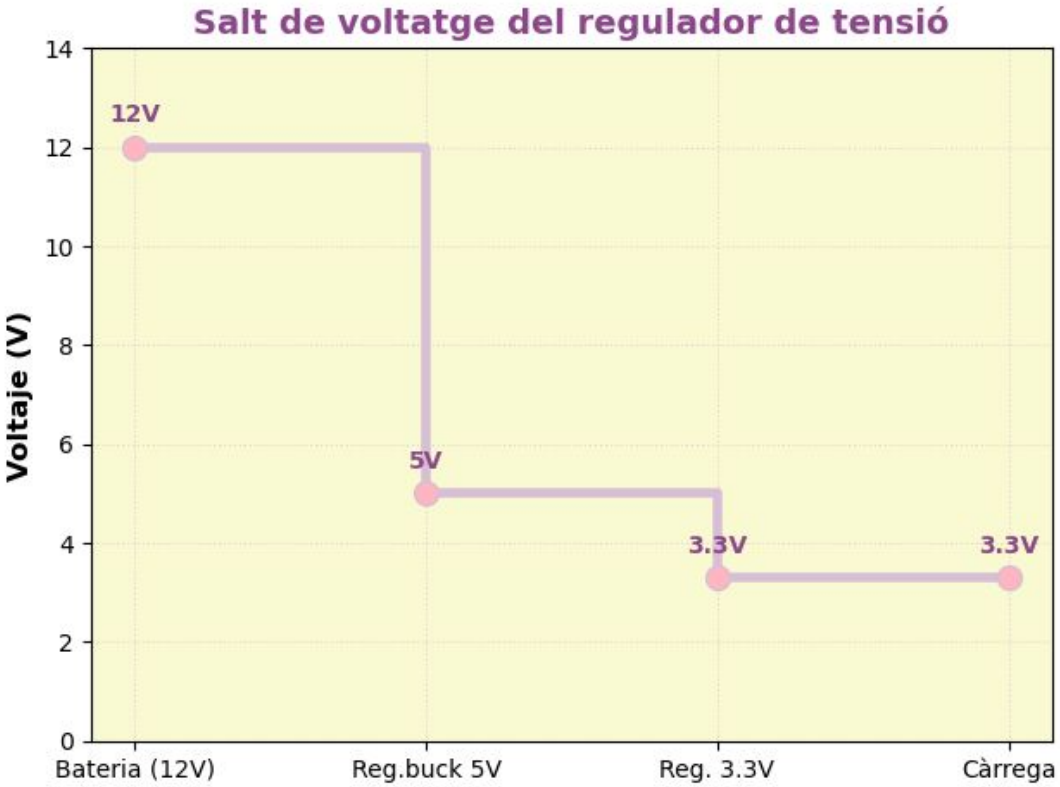


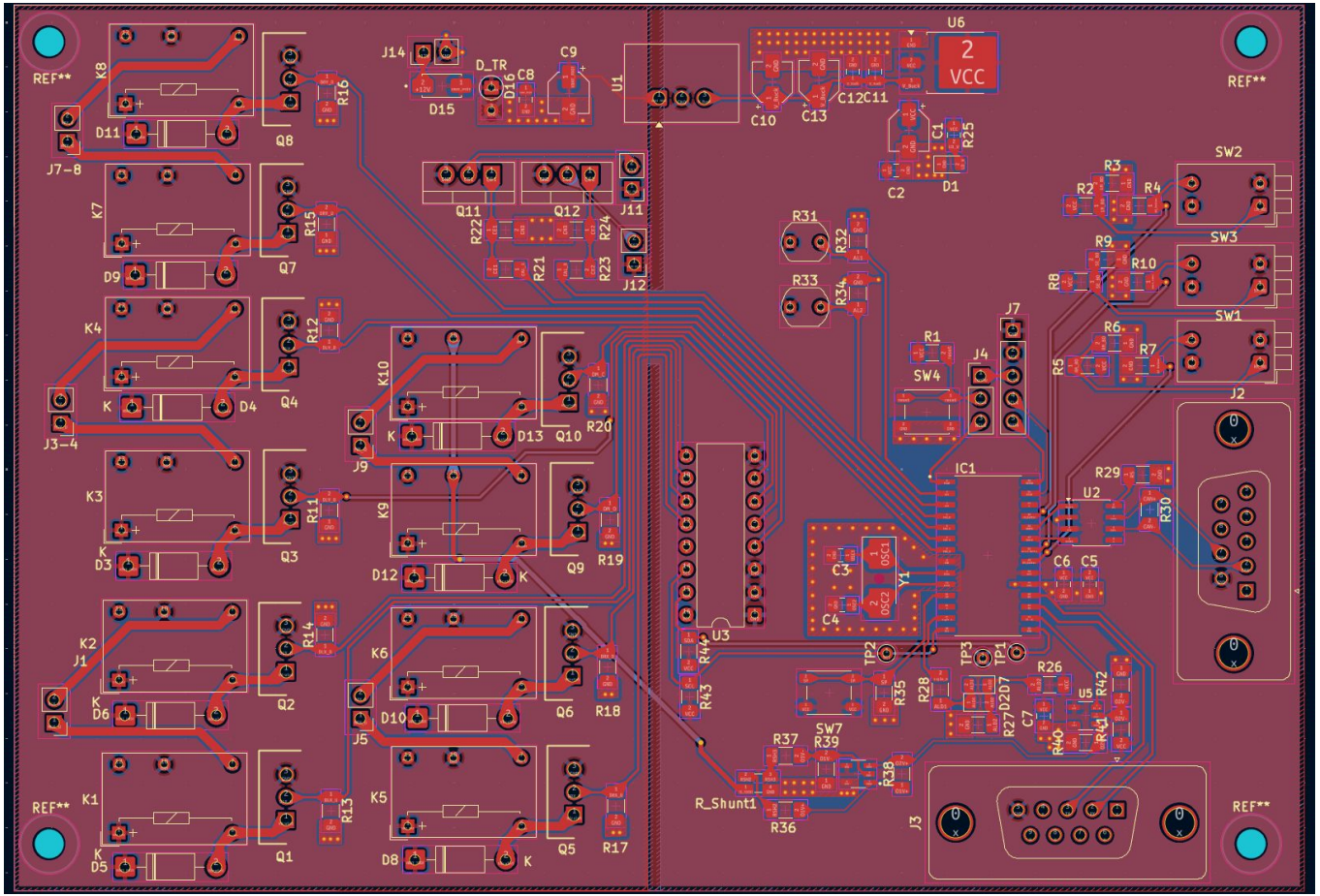
SIMULACIÓ: RELÉS





SIMULACIÓ: PART ALIMENTACIÓ, BUCK I REGULADOR DE TENSIÓ

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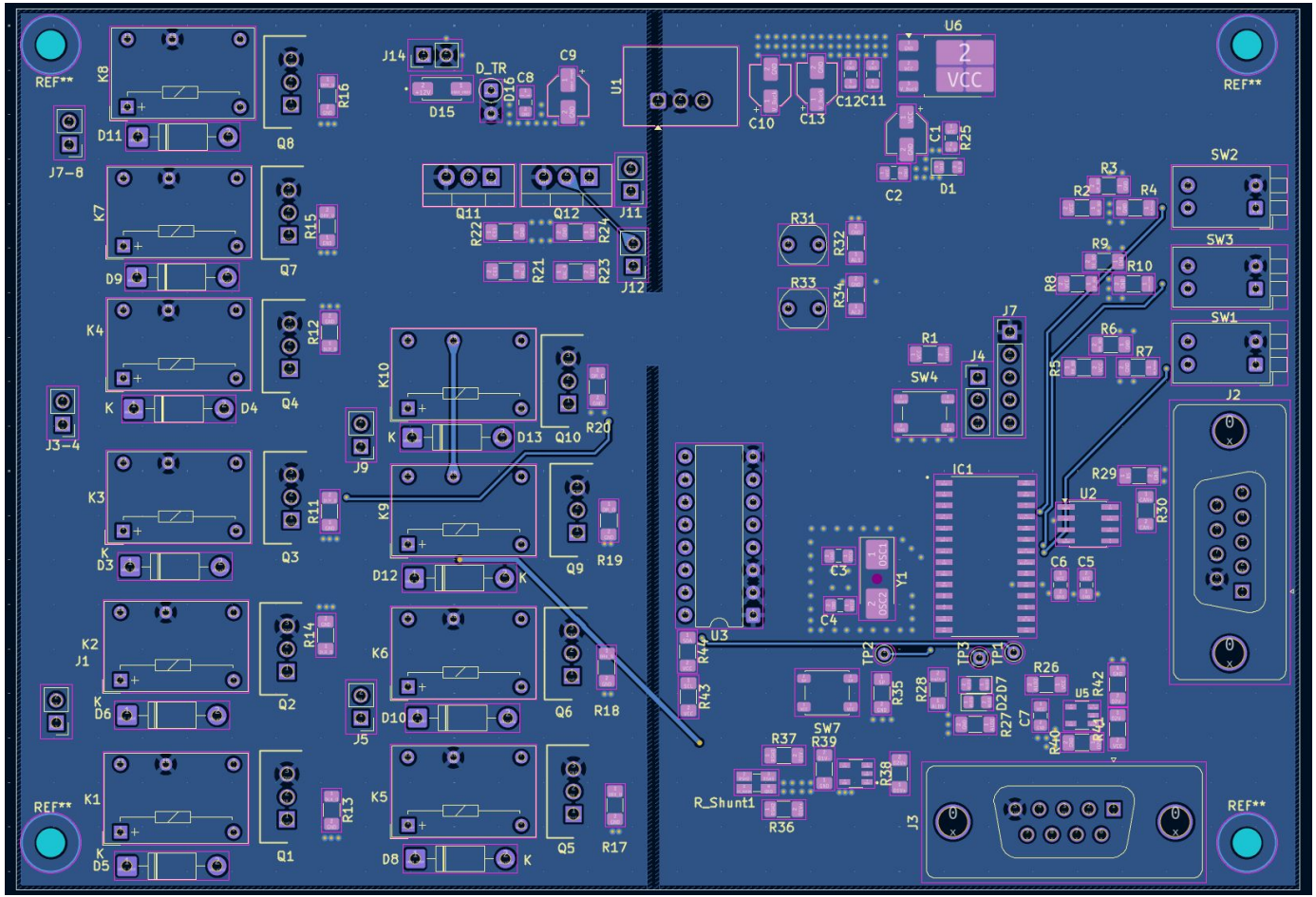
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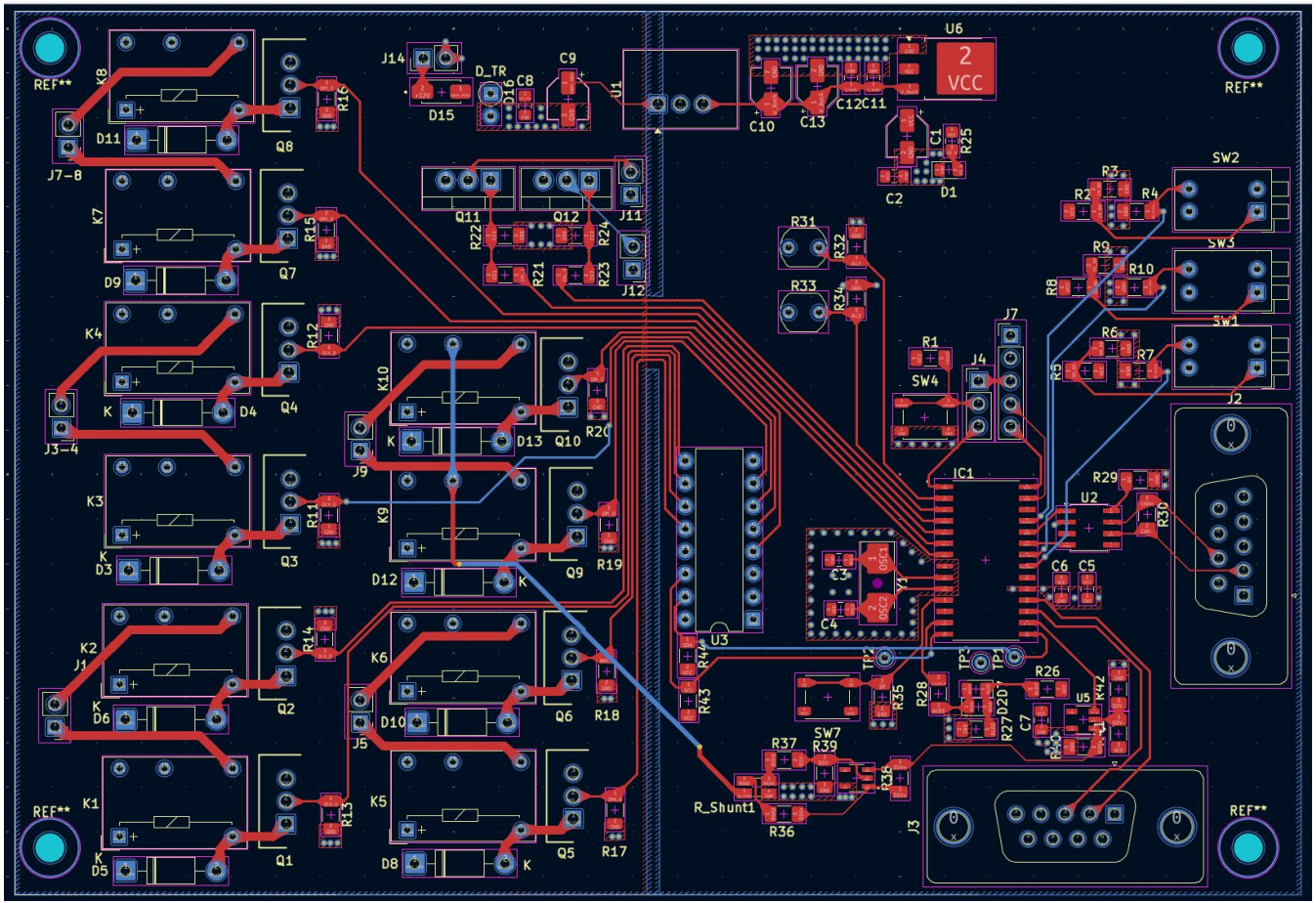
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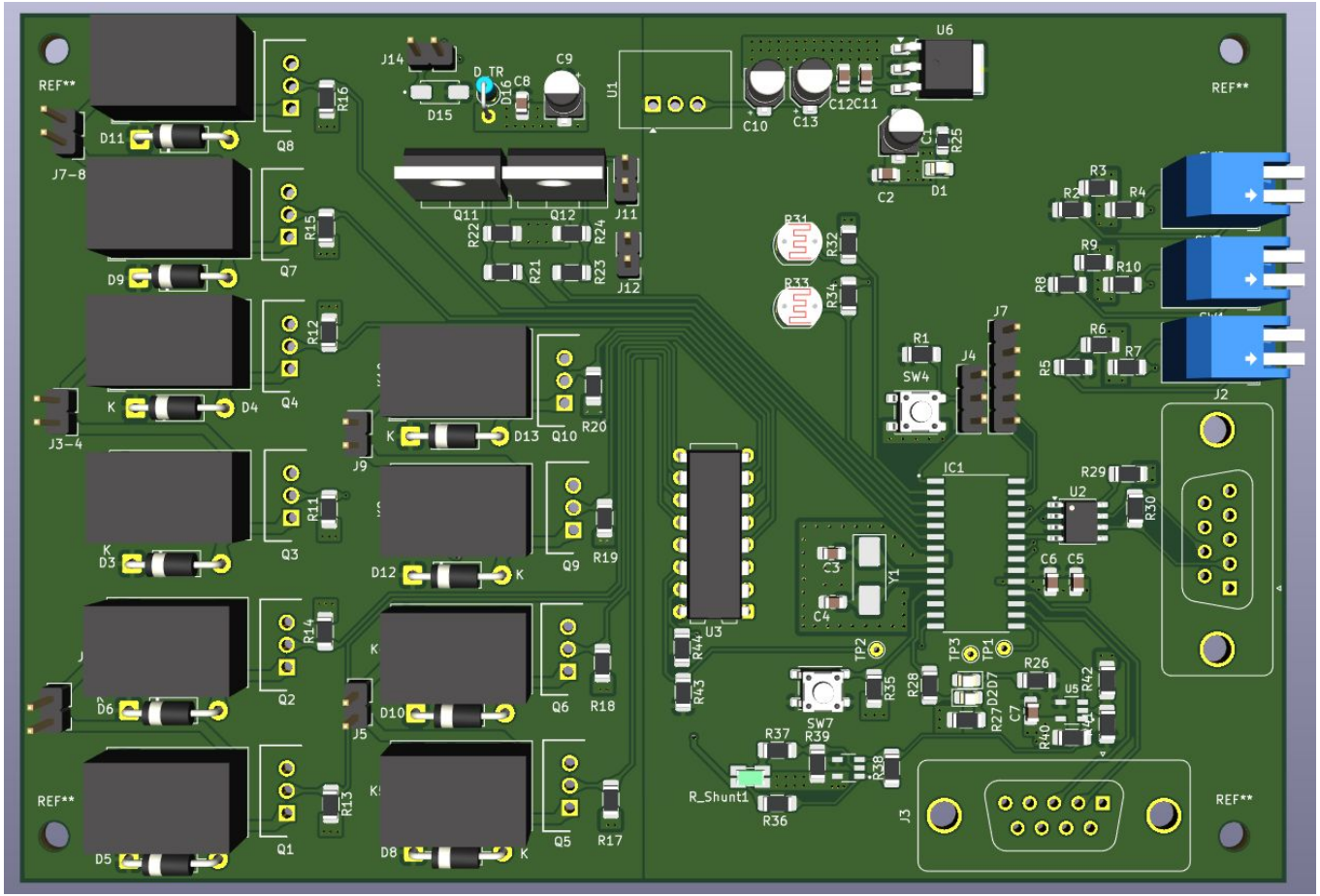
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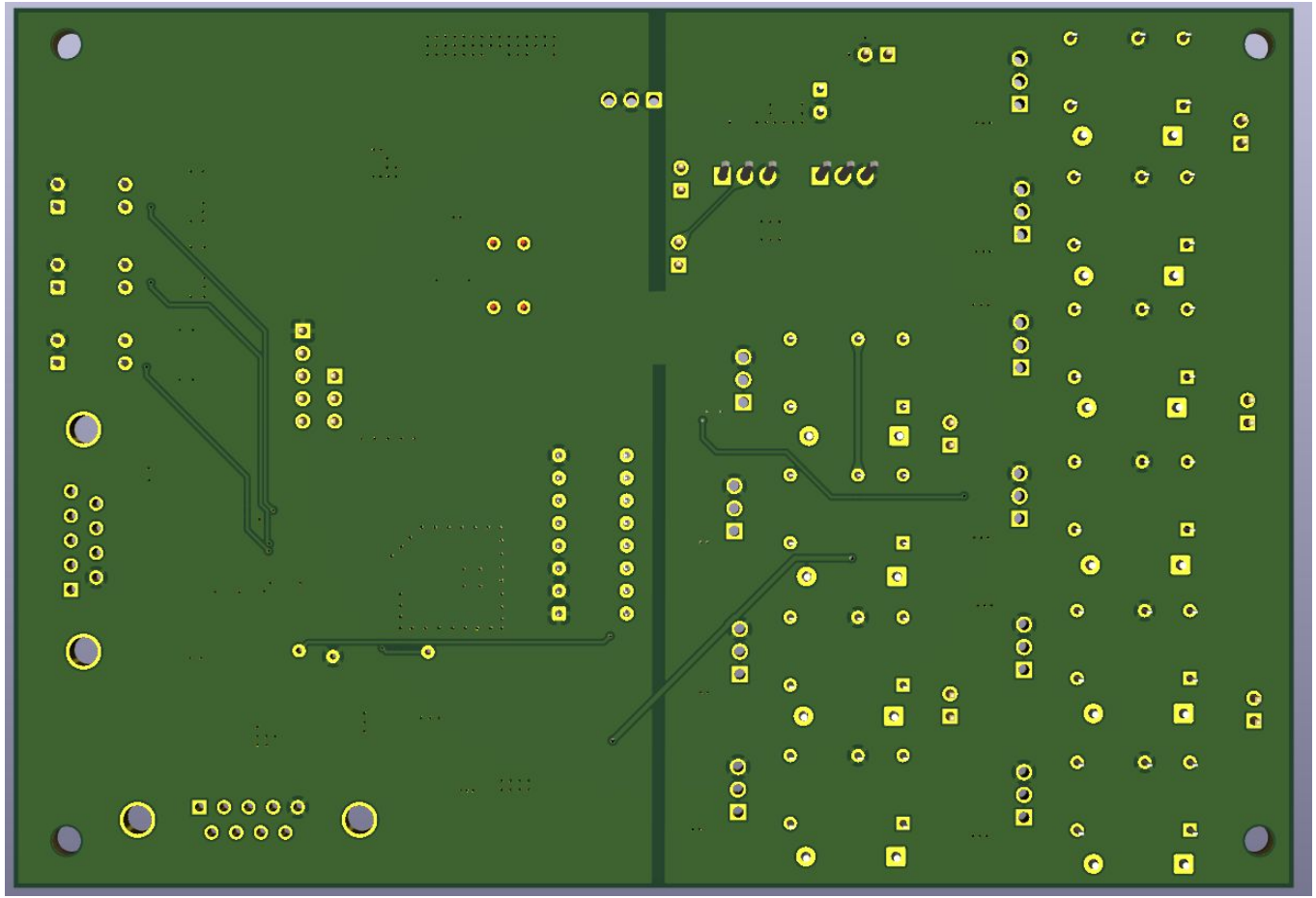
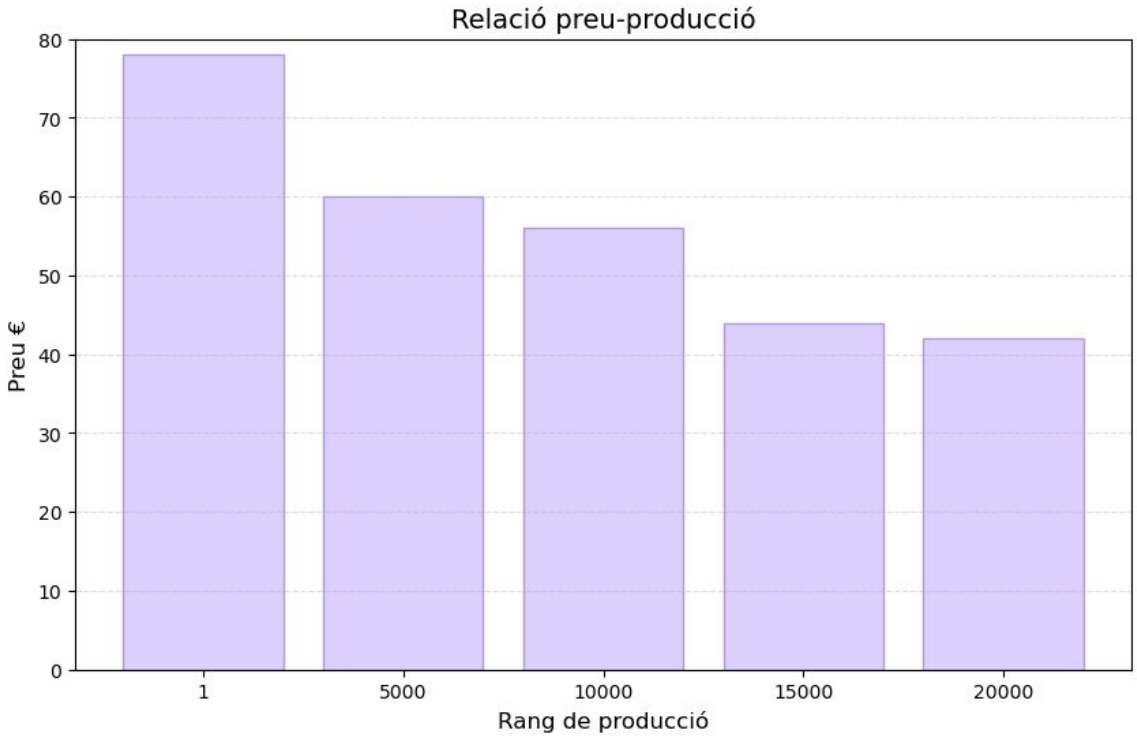


PREU PELS COMPONENTS:

	x1 PLACA	x10 PLAQUES	x50 PLAQUES	x1000 PLAQUES	x20000 PLAQUES
PREU TOTAL COMPONENTS (€)	67,47	625,04	2876,27	42681,19	842069,8

PREU PELS COMPONENTS I PLACA: TOTAL

	x1 PLACA	x10 PLAQUES	x50 PLAQUES	x1000 PLAQUES	x20000 PLAQUES
PREU TOTAL COMPONENTS (€)	67,47	625,04	2876,27	42681,19	842069,8
PREU ELABORACIÓ PLACA (€)	11,50	14,50	37	390,41	7028,52
PREU PER PLACA (€)	78,97	63,95	58,27	44,07	42,45
PREU TOTAL (€)	**78,97**	**639,54**	**2913,27**	**44071,6**	**849098,32**

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